Nexus AI: A Local Language Model Bound to an Exact Reversible Compiler

From one reversible reader to a five-provider vacuum linker over a 131,072-state symmetry sector

Driven by Dean A. Kulik

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Abstract

A language model reads text and proposes. It does not hold an exact state, and it has no way home. This paper reports the NEXUS AI series, a local system in which a small language model, Gemma 4 E4B running in LM Studio on a consumer graphics card, states what is wanted, and a deterministic reversible compiler built on the native partner ring owns everything that must be exact: the state, the route, the inverse and the status. Over twenty releases between 1 and 3 November 2026 the model's freedom narrowed to one typed call with at most two arguments, while the compiler grew from a single reversible reader into a coupled pair of readers, a goal resolver with four typed outcomes, a registry of sealed return certificates with symmetry lifting and proof composition, a stabilizer-reduced pair sector of exactly 2^{17} states mapped completely, and a vacuum linker that answers every vacuum-return goal in that sector from five primitive providers without executing a step. The sector carries laws its atlas did not state. A one-bit ledger flips on every tick, so every orbit is even and every closure's owner fixes the parity of its clock. Sixteen carrier symmetries commute with the tick. Ten long orbit families each close at their midpoint under exactly one of them. The vacuum families are owned by the ring's own carry classes. One link remains: the first meeting of a state with its own symmetry image, read from the initial boundary. The same link is missing in the single reader's two-lap closure. The architecture and its interface ledger point to a local agent whose memory is a small set of verified providers closed under symmetry and composition.

> **Status words. STANDS**: proved here, or established earlier in this series or in the literature. **RUNS**: an exact computation, exhaustive over the stated range, or landing where a sealed route placed it. **BOUND**: one provider owns the result. **OPEN**: stated, with its missing link named. **PULL**: a direction the results point to, not yet routed. The compiler itself returns four binding outcomes, defined in Section 3.3: BOUND, AMBIGUOUS, UNBOUND and UNLINKED.

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1. Introduction

Transformations constitute reality. The native partner ring is not a model laid over a domain that exists apart from it; it is the fixed space of every binary configuration and of every legal relation between configurations, and each closure is a route through it [6]. The ring does not change. The reader moves, and what the reader keeps is the record of the transitions that carried it. An artificial intelligence built on that premise does not need to hold knowledge as data. It needs to hold routes, the ways home that come with them, and a reader that can ask for them.

A language model is a reader too, and its place in the ring is exact. A one-bit reader of n-bit words is a truth table of 2^n bits, so the ring R_{2^n} holds every such reader as one of its words, and a read event, a reader together with its input, is one word of R_{2^n+n}. At n = 2 the input 01 sits at address 2 of R_2. The XOR reader is the table 0110 at address 5 of R_4, the AND reader is 0001 at address 1, and the two read events 011001 and 000101 sit at addresses 21 and 5 of R_6; splitting either event returns the reader and the input exactly. The interpretation of a word is not inside the word. It is another relation, with its own address, one ring up. That fixes the division of labour this paper reports: the language model names what is wanted, and the compiler supplies where it is, how to reach it and how to return.

Language models are fluent and inexact. Asked to step a reversible machine, a model produces a plausible state; asked for a period, it produces a plausible number. The NEXUS AI series removes both by construction. The model never computes a transition. It issues one typed tool call. A deterministic compiler holds the live state, executes or retrieves the route, verifies the way home and returns a structured result with a status. A goal the compiler cannot own comes back as a typed outcome instead of an invented answer.

The paper makes six contributions.

- An architecture in which a local model of about four billion effective parameters, on an RTX 4060 with 8 GB of video memory, operates an exact reversible compiler through typed calls, with a hash-chained ledger, read-only goal resolution and four binding outcomes (Sections 3 and 4).
- A release series of twenty versions in which each release replaced a search with a route: transport, inverse, goal binding, an event-driven first vacuum, a checksum for displacement, sealed certificates, symmetry lifting, proof composition, a stabilizer quotient, a complete sector atlas and a five-provider linker (Section 4).
- An interface ledger: every failure on the model's side was an addition to the goal, each met a typed outcome, and each was closed by narrowing the contract until only intent remained free (Section 5).
- The complete structure of the fixed-charge pair sector: the e_R ledger, sixteen commuting symmetries, ten long orbit families with one half-turn owner each, a 7,003-point quotient, vacuum families owned by the carry classes, and a parity law for every clock (Section 6).
- A separation of what is new from what the published record already holds (Section 7).
- The open link, named once, and what the series leads to (Sections 8 and 9).

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2. The native machine

2.1 Partner and ring

Write J for the reversal and C for the complement of an n-bit word, and $u = 2^n - 1$ for the all-ones word. The native partner is P(x) = J(x) when J(x) ≠ x, and P(x) = C(x) when x is a palindrome. P is an involution with no fixed point. The ring R_n lists every n-bit word once: the smaller member of each partner pair ascends on one face, its partner faces it on the other, $R_n[N - 1 - i] = P(R_n[i])$ with $N = 2^n$, and the two words at the seam are exchanged [3, 4]. The ring is addressed in O(n) operations in both directions and is never stored [4, 7].

2.2 The reader

A reader is a position p, which is a word, a carry w and a field D of retained words. One tick reads the cell under the reader, forms the difference, writes the partner there and moves by the difference:

$$x = D[p] \oplus w, \quad w' = P(x), \quad D[p] \leftarrow D[p] \oplus w', \quad p' = p \oplus x.$$

Because P is an involution the tick inverts exactly: x = P(w′), p = p′ ⊕ x, D[p] ← D[p] ⊕ w′, w = D[p] ⊕ x. From an empty field a reader never leaves the carrier $p_0 \oplus \mathrm{span}\{u, w_0, J(w_0)\}$. The carrier has two, four or eight words according to how the carry meets its mirror, and the three classes close in 6, 132 and 43,155,322 ticks at every width; they are the classes R1, R2 and R3 [1, 2]. The R3 closure returns its field to the mold at the exact half period, 21,577,661 ticks, with the reader on the complement of its start, and closes at twice that: two laps home [2, 5]. **STANDS.**

2.3 Transport and the lazy frame

Translation by a word h moves the reader and the whole field: $H_h(p, w, D) = (p \oplus h, w, D_h)$ with $D_h[q] = D[q \oplus h]$. The cell under the translated reader holds what the cell under the original held, so the difference and every later write are unchanged:

$$F\,H_h = H_h\,F, \quad H_h^2 = I, \quad H_a\,H_b = H_{a \oplus b}.$$

The computation depends on relations, not on placement. Reading the field through a frame, E[r] = V[r ⊕ s], the tick becomes x = V[s] ⊕ w, w′ = P(x), V[s] ← V[s] ⊕ w′, s′ = s ⊕ x, p′ = p ⊕ x. One cell is read and written, no key moves, and a = p ⊕ s is invariant under F while H_h shifts it by h: execution keeps the frame and transport changes it. This lazy-frame engine entered the published record as the multi-reader v3 engine [1]. **STANDS.**

2.4 A predecessor without search

The relative tick inverts from its destination alone. Given the destination (w′, E′), the displacement is x = P(w′), the preceding field is E[r] = E′[r ⊕ x] ⊕ [r = 0] w′, and the preceding carry is w = x ⊕ E[0]. A required placement binds the same way: to hold the same WHAT at the WHERE p′, the translation is h = p ⊕ p′. No candidate is generated and no trajectory is scanned. **STANDS**; 1,000 randomized cases at widths 2 to 10 return exactly (**RUNS**).

2.5 Two readers on one field

Two readers A and B share one field and tick together. Each forms its difference from the field as it stands, both write, and both move; when they stand on one cell, both writes enter it. The step factors into four

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involutions and inverts exactly for any number of readers [1]. On the eight-word carrier K = span{e_R, e_L, u} at even width, where e_R and e_L are the two central bits, the partner is affine:

$$P(x) = \Phi_A(x) \oplus u, \quad \Phi_g(x) = x \oplus t_g\, \ell(x).$$

Here ℓ(x) = 1 exactly when x is not a palindrome, which on K means that its two central bits differ, and the toggles are t_I = 0, t_J = s, t_A = u ⊕ s and t_{Ao} = u, with s = e_R ⊕ e_L. The four maps Φ_g form the Klein four-group V_4 = {I, J, A, Ao}, the centralizer of the linear part of P in GL(3, 2). The three-bit charge Q = B_D ⊕ $\Phi_A(p_A \oplus p_B)$, with B_D the XOR of the field, is conserved, so whenever the field is empty the readers stand at their original separation. Translations by K, the Klein four-group and the exchange of readers generate the 64-element equivariance group G_{64} of the two-reader machine [1]. **STANDS.**

2.6 The documented seed

At width 4 the carrier words are e_R = 0010, e_L = 0100, s = 0110, u ⊕ s = 1001 and u = 1111. Seed reader A at (0000, 0010) and reader B at (1111, 1011) on an empty field. After 806 ticks the field is empty again, both carries are back, and both readers have moved by the same word: A stands at (1001, 0010) and B at (0110, 1011). The pair has carried itself:

$$F^{806}(S) = H_{1001}(S), \quad F^{1612}(S) = H_{1001}(H_{1001}(S)) = S.$$

In equivariant dynamics this is a relative periodic orbit: the internal state closes while the placement accumulates a group element [8]. Established dynamics holds the same path at a different location; here the group is the carrier's own translations and the closure is exact. The seed is the 1,612-tick two-lap closure of the 64-seed census, whose half-period involution is the translation by u ⊕ s [1]. **RUNS.**

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3. Architecture

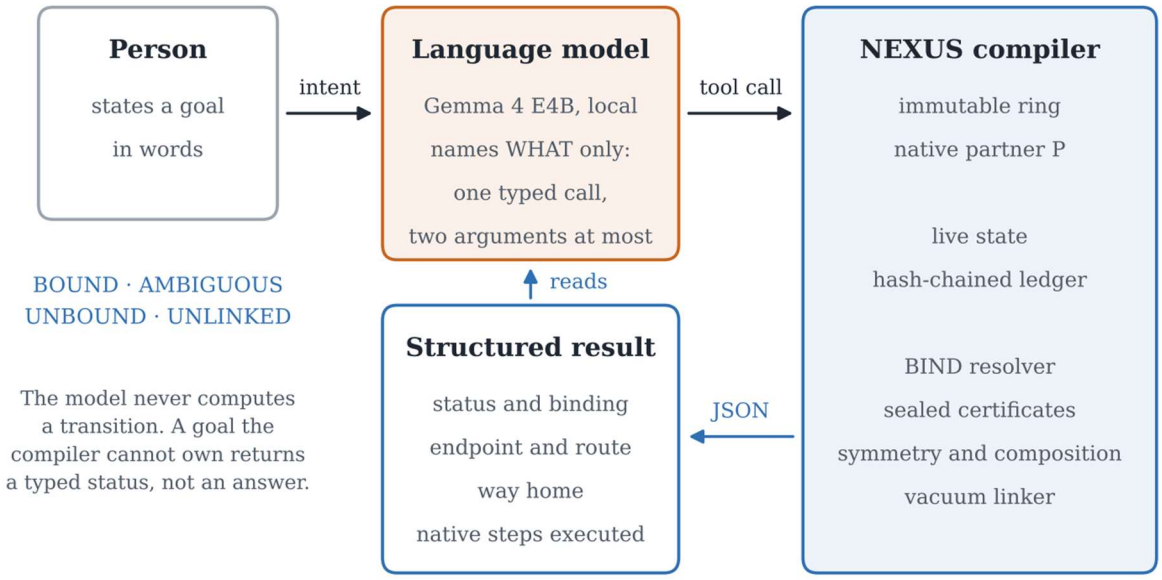


Figure 1. The division of ownership. The person states a goal; the model turns it into one typed call; the compiler owns the state, the route, the inverse and the status and returns a structured result, which the model reads back.

3.1 Components

The system runs on one desktop machine: an RTX 4060 with 8 GB of video memory, 64 GB of system memory and NVMe storage. LM Studio serves Gemma 4 E4B, a four-bit quantized build of about six gigabytes with tool calling, through an OpenAI-compatible endpoint on the local host [9, 10]. One Jupyter initialization cell installs the compiler, written in Python with the standard library only; an optional second cell connects the model. Nothing leaves the machine.

3.2 Division of ownership

The model owns one decision: which goal to request, in which typed arguments. The compiler owns everything that must be exact: the live state, the ledger, the route, its inverse, the endpoint and the status. A tool result is a JSON object with fixed fields: the status, the binding, the source and endpoint snapshots, the route, the way home, the number of native steps executed, and flags recording that the live machine and the ledger are unchanged. A missing piece is reported in a `missing_link` field and is never filled by convention.

3.3 Binding outcomes

Goal resolution follows a dependency-injection discipline. A goal is **BOUND** when exactly one provider owns it: a derived route, a verified certificate, or a composition of either with a symmetry. It is **AMBIGUOUS** when more than one provider owns it and an independent qualifier is needed. It is **UNBOUND** when no provider can exist because the goal violates an invariant, as a release from the eight-word carrier does. It is **UNLINKED** when the goal is lawful but no provider is registered. The four outcomes keep a true statement, a missing proof and an impossible request apart, and the model reports whichever one it receives.

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3.4 Ledger, seals and certificates

Every state-changing operation appends an event to a JSONL ledger whose entries are chained by SHA-256. A new release replays the old ledger, including its legacy events, before it accepts new ones. A certificate is a verified execution witness keyed to the complete source state, positions, carries and field, together with its depth, endpoint, owner symmetry and inverse, and the registry rejects a tampered record. Predictions and provider tables are sealed with SHA-256 before the readout they will be checked against (Appendix A).

3.5 Verification and roles

Each release re-runs every earlier suite before it adds its own. The 30,844 deterministic engine conditions, 10,803 BIND checks, 25,204 pair-compiler checks and 13,489 boundary assertions all carry forward into v6.0. Saved model sessions are audited from the notebook's own output, and the compiler is re-executed in an isolated process. The author directed the program: he chose each next target, ran every notebook on the machine above and routed results between systems. Several language-model systems took distinct roles outside the running system. GPT wrote and audited the notebooks, Gemini reviewed the interface, and Claude rebuilt the v5.9 sector independently from its definitions. Gemma is the only model inside the system.

4. The release series

Release	What the compiler gained	What the model sends	Checks reported
v1	Reversible reader as tools: state, step, reverse; exact return of reader, carry and field	no arguments	engine conditions
v2	Transport H_h; commuting square on a populated field	h	30,844 engine conditions
v3	Lazy frame: constant-time step and transport; earlier ledger replayed	as v2	differential runs against v2
v4	BIND: target WHERE, predecessor, owner of a target	four optional fields	10,803 BIND checks
v4.1	Contract fixed: target WHERE required; separate typed tools	target_where	30,844 and 10,803
v5	Paired-reader compiler; verified 806-tick translated return	ticks	25,204 pair checks
v5.1	First vacuum by event, no supplied depth; switch-tape inversion	none	1,612 states rebuilt, 0 mismatches
v5.2	Switch-boundary compiler: displacement from one checksum	none	every even width 4–16
v5.3	Boundary goals: the charge derives the vacuum separation	placement, target delta	13,489 assertions
v5.3.1	Target delta removed from the model and derived	placement, target WHERE	regression suite
v5.3.2	Certificate registry keyed to the complete source; AMBIGUOUS; tamper rejection	goal	registry tests

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Release	What the compiler gained	What the model sends	Checks reported
v5.3.3	Symmetry-lifted certificates for translated and exchanged copies	goal	2,676 checks
v5.4	One unified goal interface among twenty tools	placement_class, target_where_A	23 interface tests
v5.5	Proof composition: $F^t = \sigma$ gives $F^{2t} = I$	return_kind	108 checks; n = 4, 6, 8
v5.6	Index of the 64 canonical seeds with an exact certificate layer	where_B, carry_B	8 seeds re-verified
v5.7	Klein-four symmetry compiler; 20 stabilized seeds; five witnesses	carrier goal	5,176 assertions
v5.7.1	Locked two-argument symmetry goal	where_B, carry_B	regression suite
v5.8	Stabilizer-reduced pair: one reader, four field orbits; 17-bit sector	where_B, carry_B	88,620 checks, 140 cases
v5.9	Complete transit atlas of the 2^{17}-state sector	none	131,072 encode and decode
v6.0	Vacuum linker: five providers, every vacuum goal, zero native steps	where_A, carry_A	32 of 32 against the atlas

Table 1. The twenty releases, 1–3 November 2026. Each release re-runs every earlier suite. Seals are listed in Appendix A.

4.1 A reversible reader under a model (v1–v3)

The first bridge exposed three tools, state, step and reverse, over the single reader at width 4. From the reader at 0000 with carry 0001 and an empty field, the model's step call produced the difference x = 0001, the carry P(0001) = 1000, the reader at 0001 and the field cell D[0000] = 1000; its reverse call recovered the reader, the carry and the empty field exactly. The model computed neither state. Transport followed as a fourth tool. From S = (0001, 1000, {0000: 1000}) the two routes H_{0101} F and F H_{0101} were executed live through the model and both landed on (1100, 0001, {0100: 0001, 0101: 1000}), with field cells relocated and not only the reader. The lazy frame of Section 2.3 then made transport and stepping constant-time and replayed the earlier ledger without rewriting it. **RUNS.**

4.2 Goals instead of instructions (v4, v4.1)

BIND turns a requirement into a route. From the saved state with the reader at 1100, carry 0001 and field {0100: 0001, 0101: 1000}, the requirement to keep this computation and observe it at WHERE 1000 binds to h = 1100 ⊕ 1000 = 0100. The destination is (1000, 0001, {0000: 0001, 0001: 1000}), the way home is H_{0100}, and the live machine is untouched. In v4 the tool accepted four optional fields, the model sent a call that supplied no valid constraint, and the compiler answered UNLINKED. Release v4.1 required the target WHERE and gave predecessor and endpoint resolution tools of their own. The model then sent exactly `{"target_where": 8}`, the compiler returned BOUND with H_{0100} and its inverse, and the model reported the route without computing it. **RUNS.**

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4.3 Two readers carry themselves (v5–v5.2)

The author's C experiment on two readers found translated returns at ticks 2, 806, 975, 3,123, 6,733 and 11,375: the coupled system returning to its own internal configuration at another address, with no transport command. Release v5 made the pair a compiler object with an exact inverse and verified the 806-tick return of Section 2.6. The model called it with `{"ticks": 806}`, and the compiler bound the endpoint with h = 1001 and two ways home, H_{1001} or 806 inverse pair steps.

Release v5.1 found the vacuum without being given the number. The model called the first-vacuum tool with no arguments; the compiler executed to the first field-vacuum event at tick 806, bound the translation and derived the 1,612-tick period by composition. It also inverted the event log. Each reader's switch tape is $q_t = w_t \oplus w_{t+1} \oplus w_{t+2}$, the cyclic operator I + S + S² on a tape of length T. Over GF(2) its kernel has dimension 0 when $3 \nmid T$ and dimension 2 per bit of the carried word when $3 \mid T$, because $x^2 + x + 1$ divides $x^T - 1$ exactly when $3 \mid T$. At T = 1,612 the whole carry history is recovered from the switch tape with no anchor, and all 1,612 joint states were rebuilt with no mismatch. **STANDS** for the kernel law, **RUNS** for the rebuild.

Release v5.2 read the displacement off the tape. On the carrier $x_t = P(w_{t+1}) = \Phi_A(w_{t+1}) \oplus u$, and every carry appears three times in the XOR of a cyclic switch tape, so

$$h = \Phi_A(q_0 \oplus q_1 \oplus \ldots \oplus q_{T-1}) \oplus (T \bmod 2)\, u.$$

One checksum gives WHERE without reconstructing a single position. The identity holds on the 806-tick half return at every even width from 4 to 16. The checksum proposes and the read and write constraints decide: a tape altered so as to keep its checksum was rejected, because it fails the coupled dynamics. **STANDS**; **RUNS** at every even width from 4 to 16.

4.4 Goals, certificates and composition (v5.3–v5.5)

Release v5.3 compiled the goal boundary before any trajectory runs. The conserved charge fixes the separation a vacuum return must have, 1111 for the documented seed, so the compiler derives it instead of asking for it. A release from the carrier is UNBOUND, because the carrier is invariant. A same-carrier return with reader A at 1001 is spatially BOUND with h = 1001, while its route stays UNLINKED until a clock owns it (v5.3.1). Release v5.3.2 added a registry of certificates keyed to the complete source state: the 806-tick witness binds the goal at WHERE 1001, a different source or endpoint stays UNLINKED, two registered routes for one goal return AMBIGUOUS, and a tampered record is rejected. Release v5.3.3 lifted certificates through the commuting symmetries, so a translated or reader-exchanged copy of a certified source inherits the certified route, endpoint and inverse without replaying the orbit (2,676 checks). Release v5.4 put the compiler behind one goal tool among twenty. The model sent `{"placement_class": "same_carrier", "target_where_A": 9}` and received BOUND from the lifted 806-tick certificate, with zero native steps executed during the bind. Release v5.5 composed proofs:

$$F^t(S) = \sigma(S),\quad \sigma^2 = I,\quad F\,\sigma = \sigma\,F \quad\Longrightarrow\quad F^{2t}(S) = \sigma(F^t(S)) = \sigma^2(S) = S.$$

The 1,612-tick exact return is derived from the 806-tick certificate without executing 1,612 steps, and native runs at widths 4, 6 and 8 land where the composition placed them (108 checks). **STANDS** for the composition, **RUNS** for its landings.

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4.5 The carrier index and its symmetry (v5.6–v5.7.1)

Fixing reader A at (0, e_R) on an empty field gives 64 canonical seeds, one for each position and carry of reader B on the carrier [1]. Release v5.6 indexed all 64 and kept the certificate layer exact. Eight seeds with short reported depths were re-verified natively as 13 certificate records, five of the seeds AMBIGUOUS between two readings of the reader labels; 24 longer reported returns await certification; 32 seeds have no self-move readout. None was filled by convention, and no finite horizon was read as an impossibility. Release v5.7 brought in the Klein four-group on the actual carrier words, extended consistently across translated cosets so that lifted displacements conjugate correctly. It identified 20 reader-exchange stabilized seeds algebraically, re-verified five supplied first-vacuum witnesses natively at depths 1, 132, 144, 2,478 and 4,875, and found an order-four macro: the identical-reader seed reaches vacuum at tick 1 and closes at tick 4. Four of the five depths are half periods of two-lap closures in the 64-seed census, of lengths 264, 288, 4,956 and 9,750 [1]. For B = (s, e_R) the witness reads

$$F^{144}(S) = \Phi_{A0}(S), \quad F^{288}(S) = S,$$

with a second description, $H_{0110}\ \Phi_{A0}\ J_{swap}$, that coincides with the first on this seed through its stabilizer $H_{0110}\ J_{swap}$: one execution history with two symmetry names. Release v5.7.1 locked the goal to two arguments. The model sent exactly `{"where_B": "s", "carry_B": "eR"}` and received BOUND, 144 ticks, zero native steps. **RUNS.**

4.6 The sector and its atlas (v5.8–v5.9)

A seed fixed by a reader-exchange symmetry carries reader B as a function of reader A for its whole history. For the documented seed the stabilizer is $\tau = H_U\ \Phi_A\ J_{swap}$: reader B stands at $\Phi_A(p_A) \oplus U = P(p_A)$ with carry $\Phi_A(w_A)$, and the field satisfies $D(P(q)) = \Phi_A(D(q))$. One reader and one field word per partner pair of cells then describe the pair exactly, with an exact inverse, collisions included; 88,620 checks across the even widths 4 to 16 and 140 stabilized cases agree with the full machine. The quotient has 18 independent bits, and the charge fixes one of them, which leaves a fixed-charge sector of exactly 2^{17} = 131,072 states. Release v5.9 packed the sector into 17 bits, built the complete permutation and read every cycle: 2,816 cycles, 32 vacuum states lying on 18 of them, and the reference cycle of length 1,612 with its first vacuum at tick 806. Its model tool takes no arguments and reports the atlas as a graph readout, not as a clock. **RUNS.**

4.7 The vacuum linker (v6.0)

Release v6.0 reduced the 32 vacuum handoffs to five. The four translations fixed by Φ_A relate the four vacuum positions, and V_4 divides the eight carries into five classes: 0000, 1111, 0110, 1001 and the four non-palindromes 0010, 0100, 1011 and 1101. Five primitive providers, one per class, each verified forward and backward, generate all 32 handoffs by exact symmetry and agree with the atlas on all 32. A vacuum goal needs only the pair (where_A, carry_A). The model sent `{"carry_A": 2, "where_A": 0}` and received BOUND: 806 ticks to reader A at 1001 with carry 0010, the exact 1,612-tick return through two handoffs, zero native steps, and the live machine and ledger unchanged. The provider table is sealed. The primitive depths themselves are linked readouts; deriving them is the open link of Section 9. **RUNS.**

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5. The interface ledger

Release	What the model sent	What the compiler returned	Contract change
v4	a bind call with no valid constraint	UNLINKED	target WHERE required; predecessor and endpoint get their own tools (v4.1)
v5	prose: BOUND read as "a unique path exists"	BOUND for the supplied depth only	BOUND defined as one provider for the stated goal
v5.3	an invented "target_delta": 0, where the true separation is 1111	UNBOUND for a lawful goal	separation removed from the model and derived from the charge (v5.3.1)
v5.7	an invented "placement_class": "any"	UNLINKED	two-argument symmetry goal (v5.7.1)
v6.0	prose: 1,612 inverse steps lead home from the 806-tick endpoint	structured result correct: 806 inverse steps	none; the structured result is the record

Table 2. Every model-side failure recorded in the saved sessions, what the compiler returned, and how the contract changed.

Every failure on the model's side was an addition: a constraint the goal did not contain, a parameter the tool did not need, or a stronger reading of a status than the status carries. No number the compiler returned was wrong, and each addition met a typed outcome instead of a fabricated success. The repair was never a larger or smarter model. It was a narrower contract: v4 accepted four optional fields, and v6.0 accepts two integers. The principle the Gemini review of v5.7.1 identified is to reduce the model's freedom to the minimum required to express intent, and to give the compiler ownership of everything that must be exact. The ledger adds one rule of its own: the model's prose is a reading of the structured result, and where the two differ the structured result stands. **STANDS** as the observed law of this interface over twenty releases.

6. The structure of the fixed-charge sector

The v5.9 atlas is a complete readout. This section reads the laws it carries. The results come from an independent rebuild written from the definitions alone and from an independent audit notebook; both reproduce the permutation, the spectrum of Figure 2 and all 32 handoffs exactly.

6.1 Coordinates

At width 4 the carrier is K = {0000, 0010, 0100, 0110, 1001, 1011, 1101, 1111}. Its eight cells fall into four partner pairs, {0000, 1111}, {0010, 0100}, {0110, 1001} and {1011, 1101}. A sector state is reader A's position $p \in K$ and carry $w \in K$, together with the field words $z_0, \ldots, z_3$ on the representatives 0000, 0010, 0110 and 1011, subject

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to the charge condition $\ell(p) \oplus \ell(z_0) \oplus \ell(z_1) \oplus \ell(z_2) \oplus \ell(z_3) = 0$. A tick reads the cell under A, writes the partner there and its Φ_A image on the partner cell, and moves:

$$x = D(p) \oplus w, \quad y = P(x),$$
$$D(p) \leftarrow D(p) \oplus y, \quad D(P(p)) \leftarrow D(P(p)) \oplus \Phi_A(y),$$
$$p \leftarrow p \oplus x, \quad w \leftarrow y.$$

A vacuum state has an empty field. The charge then forces $\ell(p) = 0$, so $p \in \{0000, 0110, 1001, 1111\}$, and there are $4 \times 8 = 32$ vacuum states.

6.2 The carrier tables

Three facts hold at every even width. On K the partner is the complement of Φ_A, $P = H_U \circ \Phi_A$ [1]. The τ-quotient pairs every cell with its own partner, because the cell reader B writes is $\Phi_A(q) \oplus u = P(q)$. In the coordinates of K the tables of P, of V_4 and of the cell pairing are the same at $n = 4, 6, \ldots, 16$, so the sector is the same permutation at every even width, and every count below belongs to the carrier and not to width 4. **STANDS**; the tables agree at every even width from 4 to 16 (**RUNS**).

6.3 The e_R ledger

Theorem 1. Let L(S) be the e_R bit of reader A's position plus the e_R bits of the four representative field words, taken mod 2. Then $L(F(S)) = L(S) \oplus 1$ for every state of the sector.

Proof. On K the partner flips the e_R bit. On a palindrome P is the complement, which flips every bit; on a non-palindrome P is the reversal, which exchanges the two central bits, and they differ there. So $e_R(P(x)) = e_R(x) \oplus 1$. The tick moves A by x, which adds $e_R(x)$ to the position bit. It adds $y = P(x)$ to the representative of the partner pair under A, as y or as $\Phi_A(y)$, and Φ_A toggles $u \oplus s$, which has no e_R bit, so the representative's e_R bit changes by $e_R(y)$ either way. The change in L is $e_R(x) \oplus e_R(x) \oplus 1 = 1$. ■

Corollaries. Every cycle of the sector has even length. A vacuum-to-vacuum handoff lasts an even or an odd number of ticks according to the e_R bit of reader A's displacement. If a state meets its own image g(S) at tick t, then $t \equiv L(g(S)) \oplus L(S) \pmod 2$: the owner of a closure fixes the parity of its clock before the clock is known. The last corollary was drawn in the GPT audit of v6.0. **STANDS**; the flip holds on all 131,072 states, the handoff rule on all 32 handoffs and the owner rule on all 112,048 states of the long families (**RUNS**). The One Inverter clock changes the weight of its word by exactly one on every tick and every one of its orbits has even length [12]. That is the same path at a different location: one bit that flips on every tick forces every closed route to be even.

6.4 The sixteen symmetries

Theorem 2. Among the maps of the sector of the form $p \mapsto g(p) \oplus k$, $w \mapsto g(w)$, with the field carried along as $D'(g(q) \oplus k) = g(D(q))$, where g is linear on K and $k \in K$, exactly sixteen commute with the tick: those with $g \in V_4$ and $k \in \{0000, 0110, 1001, 1111\}$.

Proof. The difference transforms as $x \mapsto g(x)$, so the written partner commutes with the map only if $g\,P = P\,g$, which places g in the centralizer V_4 of the linear part of P and fixes u. The partner pairing of cells survives only

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if $P(g(q) \oplus k) = g(P(q)) \oplus k$, that is, $\Phi_A(k) = k$, and on K this holds exactly when $\ell(k) = 0$, which also preserves the charge. Moves and writes then commute term by term. ■

The reader exchange is $A \cdot H_{1111}$. The sixteen are the elements of G_{64} that commute with τ, taken modulo τ itself: an element (g, k, sw) commutes with $\tau = (A, u, 1)$ exactly when $\Phi_A(k) = k$, which leaves $4 \times 4 \times 2 = 32$ elements, and τ acts as the identity on the sector. Of all 1,344 affine maps of the carrier, exactly these sixteen commute with the tick on all 131,072 states. **STANDS**; exhaustive (**RUNS**).

6.5 Ten long families and their owners

Ten cycle lengths occur exactly eight times each. Each such family is one orbit of the sixteen symmetries, and each of its cycles is carried onto itself by exactly one non-identity symmetry, which acts on it as the half-turn: $F^{T/2}(S) = g(S)$. By the composition of Section 4.4 the period is twice the first meeting with that image. No earlier meeting with any image is possible, because a symmetry that carries a state of the cycle onto the cycle belongs to its stabilizer. The composition that v5.5 applied to one certificate therefore holds for every long orbit of the sector, and each family has one owner. **BOUND**: one owner per family, read from the complete map. The owner of the 806-tick half return, H_{1001}, is the translation the switch checksum of v5.2 produces, $h = u \oplus e_R \oplus e_L$. The 64-seed census holds the same law at a different location: every short two-lap closure there reaches vacuum at its half period under a fixed-point-free involution of G_{64} [1]. Here the law holds for every long orbit of the sector, whether it visits the vacuum or not.

Cycle length T	Half T/2	Half-turn owner g	$L(g(S)) \oplus L(S) = T/2 \bmod 2$
30	15	$A \cdot H_{1111}$, the reader exchange	1
94	47	H_{1111}	1
98	49	$A \cdot H_{0110}$	1
132	66	A_0	0
212	106	A	0
434	217	$A \cdot H_{1111}$, the reader exchange	1
1,612	806	H_{1001}	0
2,260	1,130	A	0
3,948	1,974	$A \cdot H_{1001}$	0
5,186	2,593	H_{1111}	1

Table 3. The ten long families. Each holds eight cycles, and on each cycle exactly one non-identity symmetry acts, as the half-turn.

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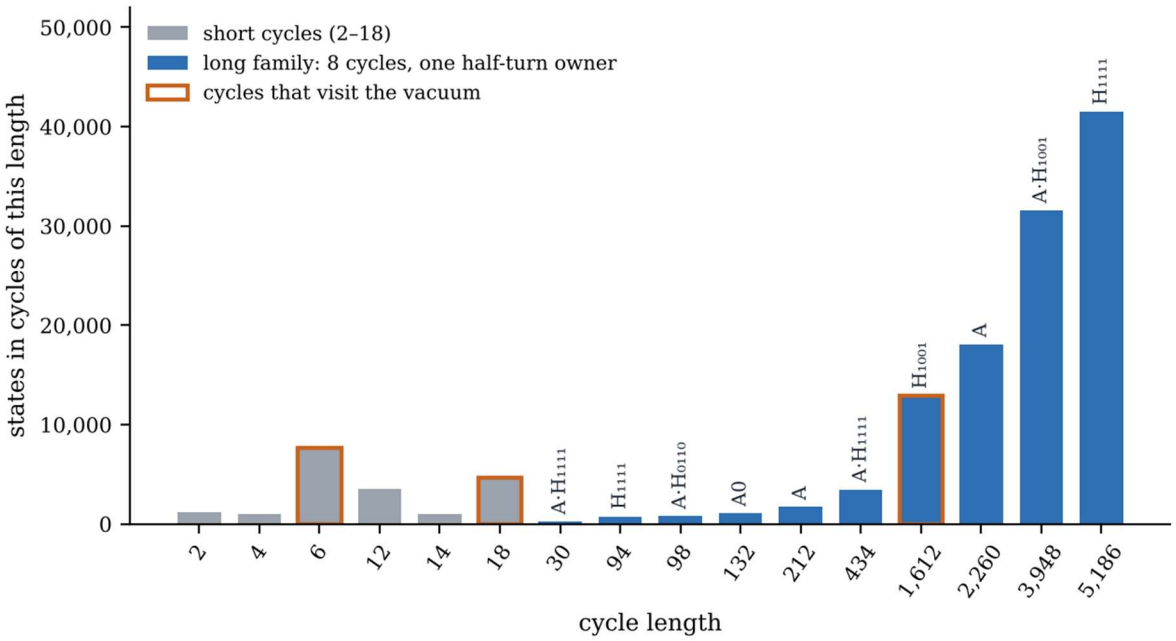


Figure 2. The 131,072 states of the fixed-charge sector by cycle length. Ten long families of eight cycles (blue) hold 112,048 states, each labelled with its half-turn owner; 2,736 short cycles of lengths 2 to 18 (grey) hold 19,024. Outlined bars are the three cycle types that visit the vacuum.

6.6 What is left: the quotient

Divide the long families by the sixteen symmetries. The group acts freely on their states, so 112,048 states become 7,003 points, and each family becomes one cycle of half its length: 15, 47, 49, 66, 106, 217, 806, 1,130, 1,974 and 2,593. The ten lengths have no common divisor, and 2,593 is prime. A random permutation of 7,003 points has $H_{7003} \approx 9.43$ cycles on average [11]; this one has ten. The quotient is where the remaining clock lives. **RUNS** for the quotient; the clock is **OPEN** (Section 9).

6.7 Vacuum families and the carry classes

Class	Carries of reader A	Pair sector: cycle, ticks between vacua	Single reader: period [2]
R1	0000, 1111	6; 2 and 1	6
R2	0110, 1001	18; 18	132
R3	0010, 0100, 1011, 1101	1,612; 806	43,155,322

Table 4. The 32 vacuum states by the carry class of reader A.

The 32 vacuum states split by the carry class of reader A, the classification the single reader already carries: R1 = {0, υ}; R2, the palindromes and anti-palindromes; R3, the rest [1, 2]. The vacuum position does not enter, and all four positions in a class behave alike. R1 keeps its single-reader period because reader B's mirror write is erased before A reaches it. From (0000, 0000), A writes 1111 under itself and B writes 1111 on the partner cell 1111; the next tick erases both, and only then does A jump to 1111. In R2 and R3 reader A reads cells that

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reader B has written, and the clocks shorten from 132 to 18 and from 43,155,322 to 1,612. The class owns the family at both locations, and the period belongs to the location. **BOUND.**

6.8 The vacuum return map

On the 32 vacuum states, V = {(p, w) : ℓ(p) = 0, field empty}, the first-return map of the sector is the map v6.0 compiles:

$$M(p, 0000) = (p, 1111) \quad \text{after } 2 \text{ ticks},$$

$$M(p, 1111) = (p \oplus 1111, 0000) \quad \text{after } 1 \text{ tick},$$

$$M(p, w) = (p, w) \quad \text{after } 18 \text{ ticks}, \quad w \in \{0110, 1001\},$$

$$M(p, w) = (p \oplus 1001, w) \quad \text{after } 806 \text{ ticks}, \quad w \in R3.$$

It is generated from five providers by the sixteen symmetries, and every handoff time has the parity Theorem 1 assigns to its displacement: even for the displacements 0000 and 1001, odd for 1111. The 32 handoffs form 18 vacuum cycles: two cycles of 6 ticks with four handoffs each, eight of 18 ticks with one, and eight of 1,612 ticks with two. **RUNS** against all 32 atlas entries.

6.9 The short cycles

The other 2,736 cycles, of lengths 2 to 18, hold 19,024 states. Their stabilizers are larger: many lie inside the fixed sets of J, A or A0, and some are free orbits of all sixteen symmetries. They are complete in the atlas, and the vacuum linker does not need them.

7. What is new

Against the published record [1–7], the following are new in this series.

- A local language model bound to an exact reversible compiler through typed goal calls, with four binding outcomes, read-only resolution, a hash-chained ledger and sealed providers. Every exact quantity is owned by the compiler and none by the model.
- BIND: goals resolved by the inverse and transport laws instead of searched, with the missing link named whenever a goal does not bind.
- Certificates as memory: verified witnesses keyed to complete source states, lifted through the commuting symmetries and composed into longer returns, answering goals with zero native steps.
- The interface law of Section 5: model-side failures are additions, they meet typed outcomes, and they close by narrowing the contract; the structured result is the record.
- The switch-tape displacement checksum, $h = \Phi_A(\bigoplus q_t) \oplus (T \bmod 2)\, u$, and its use as a boundary test.
- The τ-reduced pair, with one reader and one field word per partner pair of cells, the 17-bit fixed-charge sector and its complete map.
- Theorem 1, the e_R ledger: every orbit of the sector is even, and every owner fixes the parity of its own clock.

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- Theorem 2, the sixteen symmetries of the sector, as V_4 with the four Φ_A-fixed translations and as the centralizer of τ in G_{64} modulo τ.
- Half-turn ownership of every long orbit with one owner per family, and the 7,003-point quotient that remains.
- Vacuum families owned by the ring's carry classes, the R1 period kept and the R2 and R3 periods shortened by crosstalk, and the five-provider vacuum linker.

The following were already published and are used here: the partner ring and its addresses [3, 4]; carrier confinement, the three classes and their periods, and the two-lap half return of the single reader [1, 2, 5]; the affine partner on the carrier, the Klein four-group as its centralizer, the conserved charge, the four-involution inverse, G_{64} and the 64-seed census with its monodromy cocycle [1]; and the lazy-frame engine [1].

8. Where it leads

8.1 Memory as generators

The sector holds 131,072 states. The linker holds five providers and a group of sixteen, and from them it returns every vacuum fact of the sector with its route and its way home, without running a step. What is stored is a shape: a generating set and the symmetries that act on it. The certificate registry, the lifting and the composition are the working parts of a memory that keeps providers and derives instances. A local assistant built this way keeps what it has verified as a small set of providers closed under symmetry and composition, and it answers by routing, not by recalling. It needs a language model to hear the request and a compiler to own the answer. It does not need endless data. **PULL.**

8.2 Capability lives in the compiler

Every exact result in the series came from the compiler. The model, with about four billion effective parameters on 8 GB of video memory, never needed to know how a route works in order to request it, and the contract narrowed monotonically, from four optional fields to two integers. The end of that trajectory is a contract whose only argument is the WHAT: the model names the goal, and the compiler binds WHERE and ROUTE. The reader-ring correspondence of Section 1 says why size does not buy exactness. A reader of n-bit words that returns m bits is one word of $R_{m \cdot 2^n}$; the model is one such word among all the others, and an exact route through an immutable structure is not a property of any one reader. A small model on consumer hardware suffices when the exact part is compiled. **PULL.**

8.3 One link closes three clocks

Three clocks in the record wait on one link. The single reader's R3 closure meets its own image at tick 21,577,661 [2]. The 64-seed cocycle composes first-vacuum depths that are read from a census [1]. The sector's quotient closes the R3 vacuum at tick 806. Each is the first meeting of a state with its own symmetry image, read from the initial boundary. In each case the owner is already bound, and Theorem 1 fixes the parity of the meeting from the owner alone. What is missing is the route that gives its magnitude. The quotient lengths do not read as an arithmetic formula, so that route runs through the dynamics: a section whose return map composes pieces the compiler already owns. **PULL**; the link itself is **OPEN**.

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8.4 Adjacent sectors and ports

The same construction applies to the sector of the other charge, to seeds stabilized by the other Klein-four owners, and to the two-shell carrier of 32 words, where the partner becomes quadratic and the charge breaks except where a shell exchange restores it [1]. Release from the eight-word carrier is UNBOUND under the isolated carrier rule, so release is a port between carriers. The next tools are ports, under the same contract discipline: typed, read-only where possible, and bound or refused by the compiler. **PULL.**

8.5 Search replaced by route

Read as a whole, the series is the reverse-the-gap procedure built as software. Each release found a place where the system searched, by stepping, by scanning or by asking the model, and replaced it with a route the structure already contained: transport for relocation, the inverse for predecessors, BIND for requirements, the vacuum event for a supplied depth, the checksum for displacement, the certificate for a replayed orbit, the symmetry for a copy, composition for a longer return, the stabilizer for the second reader, the atlas for a scan, and five providers for thirty-two handoffs. The ring is fixed; the route is what the system learned to read. A language model attached to an immutable structure becomes a reader of routes and stops being a generator of candidates. **STANDS** as a description of the series.

9. Frontier

The open links are named here once. First, the first-meeting time of a state with its own symmetry image, read from the initial boundary: 806 in the sector, 21,577,661 for the single reader, and the first-vacuum table of the 64-seed census. Second, release across carriers, UNBOUND under the isolated carrier rule and waiting on a port law. Third, vacuum monodromy outside the τ-stabilized sector. Fourth, commuting maps of the sector outside the affine family, which Theorem 2 does not cover (Technical note A). Each is a missing link in the chain, not a missing value in the data. **OPEN.**

Technical notes

> **Technical note A (Claude, independent rebuild of the sector)**
>
> Theorem 2 classifies the commuting maps of affine form on the carrier, and the exhaustive check covers all 1,344 of them. Commuting maps that are not affine on the carrier are excluded by neither. The cycle multiplicities of Section 6.5 are accounted for by the sixteen; any larger group would have to preserve them.
>
> **Technical note B (GPT, review of v5.7.1)**
>
> Calling a certificate immutable requires content-addressed identity, binding to the complete source state, deterministic re-verification and protection against substitution. The SHA-256 seals and source fingerprints supply the first three. Tamper resistance of a stored registry is a property of its implementation, and it is distinct from the immutability of the ring, which the certificate only records.

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Data and code

- `NEXUS_AI_v6_0_Unified.ipynb`: the complete v6.0 compiler with every earlier suite, the vacuum linker and an optional LM Studio cell (Python 3.11, standard library).
- `NEXUS_Independent_Atlas_Audit.ipynb`: one cell that rebuilds the sector from the carrier definitions and checks Theorems 1 and 2, the ten owners, the quotient and the 32 handoffs (standard library, under ten seconds).
- `nexus_v59_rebuild.py`: the independent rebuild with the width check of Section 6.2, the carry classes, the R1 trace and the parity of every clock (standard library, about fifteen seconds).

To rerun, place each file in its own empty directory and run it with an isolated interpreter (`python3 -I`).

Appendix A. Seals and digests

Object	SHA-256
v5.2 switch-boundary compiler, pre-readout seal	`261d35de94944a60bf1fbffd6019513734f567c50f54945cb60cca9f0a8036d0`
v5.3 boundary goal compiler, pre-readout seal	`d209108b4a9c4e8cbe96d0addf590d02ac194e110cd0aa7fe872469f51e209d5`
v5.9 transit atlas, prediction seal	`54a23c4b2bf6e6db6a4322ec11361f36bcc9ce5aea8c13e8bc7647e4b0d55a60`
v6.0 primitive provider table	`8075c0444c43a239be2da515e926695d83d6157a445e3f18ae970065666658a7`
Ledger chain head, v6.0 run	`d3c28760a59dc2aa6f4de437de31d217e0e9e976af823b3c183a047cb1513077`

Appendix B. Model calls in the saved sessions

Release	Call sent by the model	Compiler result
v4.1	`nexus_bind {"target_where": 8}`	BOUND; H_{0100}; live state unchanged
v5	`nexus_pair_bind_return {"ticks": 806}`	BOUND; A at 1001, B at 0110, field empty
v5.1	first-vacuum tool, no arguments	first vacuum at 806; h = 1001; period 1,612
v5.3.2	`nexus_pair_bind_goal_from_certificate`, two goals	BOUND, 806 ticks; UNBOUND outside the carrier
v5.4	`nexus_pair_capabilities {}`, then `nexus_pair_goal {"placement_class": "same_carrier", "target_where_A": 9}`	BOUND; zero native steps
v5.7.1	`nexus_pair_carrier_symmetry_goal {"where_B": "s", "carry_B": "eR"}`	BOUND; 144 ticks under Φ_{A0}
v5.8	the same call	BOUND; stabilizer H_{0110} J_{swap}; four field orbits
v5.9	`nexus_pair_reduced_atlas {}`	atlas readout; 806 reference
v6.0	`nexus_pair_vacuum_goal {"carry_A": 2, "where_A": 0}`	BOUND; 806 ticks; A at 1001; 1,612 exact return

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The NEXUS AI notebooks were written and audited with GPT (OpenAI), Gemini (Google) reviewed the interface, and Claude (Anthropic) rebuilt the fixed-charge sector independently and drafted this paper with the author. Gemma 4 E4B ran locally as the model inside the system.

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